

2010 版

一. 证:

$$(1) d_3(\lambda) = \lambda^2(\lambda-2)^2(\lambda+1)^3$$

$$d_2(\lambda) = \lambda(\lambda-2)$$

$$d_1(\lambda) = 1$$

故 Smith 标准形为:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \lambda(\lambda-2) & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda^2(\lambda-2)^2(\lambda+1)^3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

b) 不相似

$$|\lambda I - A| = (\lambda - a)^n$$

$$|\lambda I - B| = (\lambda - a)^n + (-1)^{n+2} \varepsilon \begin{vmatrix} \lambda - a & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \dots & \dots & \lambda - a \end{vmatrix}$$

$$= (\lambda - a)^n - \varepsilon$$

故 A, B 特征值不同 故不相似

二. 由已知 $A = \begin{bmatrix} 1 & 0 & 3i \\ 0 & 4 & 0 \\ -3i & 0 & 1 \end{bmatrix}$

$$|A - \lambda E| = (\lambda - 4)(4 - \lambda)(\lambda + 2)$$

当 $\lambda = 4$ 时, $A - 4E = \begin{bmatrix} -3 & 0 & 3i \\ 0 & 0 & 0 \\ -3i & 0 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -i \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$\alpha_1 = (i, 0, 1)^T, \alpha_2 = (0, 1, 0)^T$$

单位化: $u_1 = (\frac{i}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})^T, u_2 = (0, 1, 0)^T$

当 $\lambda = -2$ 时, $A + 2E = \begin{bmatrix} 3 & 0 & 3i \\ 0 & 6 & 0 \\ -3i & 0 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & i \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$\therefore \alpha_3 = (-i, 0, 1)^T \text{ 单位化 } u_3 = (-\frac{i}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})^T$$

故 $U = (u_1, u_2, u_3) = \begin{bmatrix} \frac{i}{\sqrt{2}} & 0 & -\frac{i}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$

则 $X = UY, Y = (y_1, y_2, y_3)^T$ 使

$$f(x) = x^H A x = Y^H (U^H A U) Y$$

$$= 4\bar{y}_1 y_1 + 4\bar{y}_2 y_2 - 2\bar{y}_3 y_3$$

三. $AA^H = \begin{bmatrix} 5 & 0 & -5 \\ 0 & 0 & 0 \\ -5 & 0 & 5 \end{bmatrix}$

$$|AA^H - \lambda E| = \lambda^2(10 - \lambda)$$

特征值为 $\sqrt{10}$

$$AA^H - 10E \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \alpha_1 = (-1, 0, 1)^T$$

$$u_1 = (-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})^T$$

$$AA^H \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \alpha_2 = (0, 1, 0)^T$$

$$\alpha_3 = (1, 0, 1)^T$$

则 $U_1 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}^T, U = \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$

令 $V_1 = A^H U, \Delta^{-1} = \begin{bmatrix} -\frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$

则 $V = \begin{bmatrix} -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix}$

故 $A = U \begin{bmatrix} \sqrt{10} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} V^H$

四. 证明:

B 是正定 H 阵 则 $B = Q^H Q$

$$|A+B| = |A+Q^H Q|$$

$$= |Q^H (Q^H)^{-1} A Q^T Q + Q^H Q|$$

$$= |Q^H| |(Q^H)^{-1} A Q^T + E| |Q|$$

$\because A$ 是半正定 H 阵 $\therefore A$ 的特征值全为非负

$$\therefore |(Q^H)^{-1} A Q^T + E| \geq |E| = 1$$

又 $A \neq 0$ 故 $|(Q^H)^{-1} A Q^T + E| > 1$

$$\therefore |A+B| > |Q^H| |Q| = |Q^H Q| = |B| \text{ 得证.}$$

五. $P(A) = \frac{1}{2}$

$$\sum_{k=0}^{\infty} (k+1)x^k = \sum_{k=0}^{\infty} (x^{k+1})' = (\sum_{k=0}^{\infty} x^{k+1})'$$

$$= (\frac{x}{1-x})', \quad |x| < 1$$

$$= \frac{1}{(1-x)^2}, \quad |x| < 1$$

$$R = 1$$

$\because P(A) < R$ 故 $\sum_{k=0}^{\infty} (k+1)A^k$ 收敛.

$$\text{且 } \sum_{k=0}^{\infty} (k+1)A^k = (E-A)^{-2} = \dots$$

六. 证:

$$|A - \lambda E| = -(\lambda + 1)^3$$

$$A + E \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank}(A+E) = 1$$

$$q = n - \text{rank}(A+E) = 2 \text{ 块.}$$

$$(A+E)^2 \sim \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank}(A+E)^2 = 0$$

阶数 ≥ 2 的 Jordan 块有 1 个

故 Jordan 标准形为 $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$

$$\varphi_A(\lambda) = (\lambda + 1)^2$$

$$p(x) = a_0 + a_1 x, \text{ 则 } p'(x) = a_1$$

$$p(1) = f(1) = a_0 + a_1$$

$$p'(1) = f'(1) = a_1$$

$$\therefore \begin{cases} a_0 = f(1) - f'(1) \\ a_1 = f'(1) \end{cases}$$

$$\therefore f(A) = [f(1) - f'(1)]E + f'(1)A$$

$$(1) f(A) = e^A$$

$$f(x) = e^{2x}, f(1) = e^2, f'(1) = e^2$$

$$\text{则 } f(A) = e^2 A$$

$$(2) f(x) = \sin x$$

$$\text{则 } f(A) = -\pi E + \pi A$$

七. 证:

$$|A - \lambda E| = -\lambda(\lambda^2 + 2)$$

$$\therefore \lambda_1 = 0, \lambda_2 = \sqrt{2}i, \lambda_3 = -\sqrt{2}i$$

$$\text{当 } \lambda_1 = 0 \text{ 时, } A - 0E \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -i \\ 0 & 0 & 0 \end{bmatrix}, \alpha_1 = (0, \frac{i}{\sqrt{2}}, \frac{1}{\sqrt{2}})^T$$

$$\text{当 } \lambda_2 = \sqrt{2}i \text{ 时, } A - \sqrt{2}iE \sim \begin{bmatrix} 1 & \sqrt{2}i & 0 \\ 0 & 1 & i \\ 0 & 0 & 0 \end{bmatrix}, \alpha_2 = (-\frac{\sqrt{2}}{2}, -\frac{i}{2}, \frac{1}{2})^T$$

$$\text{当 } \lambda_3 = -\sqrt{2}i \text{ 时, } A + \sqrt{2}iE \sim \downarrow$$

$$G_1 = \alpha_1 \alpha_1^H$$

$$G_2 = \alpha_2 \alpha_2^H$$

$$G_3 = \alpha_3 \alpha_3^H$$

$$\text{则 } A = 0 \cdot G_1 + \sqrt{2}i G_2 - \sqrt{2}i G_3$$

$$18. \text{ 令 } \begin{bmatrix} 2x_1 - x_2 \\ 3x_2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = AX$$

$$\text{则 } \|x\| = \|Ax\|_2$$

$$\textcircled{1} \text{ 非负性. 当 } x \neq 0 \text{ 时, } \|x\| = \|Ax\|_2 > 0$$

由 A 是满秩阵.

$$Ax = 0 \text{ 只有零解.}$$

$$\therefore \text{当 } x \neq 0 \text{ 时, } \|x\| > 0$$

$$\textcircled{2} \text{ 齐次性. } \|kx\| = \|kAx\|_2 = |k| \|Ax\|_2 = |k| \|x\|$$

③ 三角不等式:

$$\|x+y\| = \|A(x+y)\|_2 = \|Ax + Ay\|_2$$

$$\leq \|Ax\|_2 + \|Ay\|_2$$

$$= \|x\| + \|y\|$$

$\therefore$ 是 C^2 上的向量范数

九. 证:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 0 & 2 \\ 2 & 4 & 6 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \\ 1 & 0 \\ 2 & 0 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$A^+ = C^H (C C^H)^{-1} (B^H B)^{-1} B^H = \begin{bmatrix} -\frac{1}{30} & \frac{1}{6} & \frac{1}{3} & -\frac{1}{15} \\ \frac{1}{15} & -\frac{2}{15} & -\frac{4}{15} & \frac{2}{15} \\ \frac{1}{30} & -\frac{1}{30} & \frac{1}{15} & \frac{1}{15} \end{bmatrix}$$

则最佳最小二乘解为 $x = A^+ b$

$$\text{求得: } x = \begin{bmatrix} \frac{1}{10} \\ \frac{1}{5} \\ \frac{2}{10} \end{bmatrix}$$